

BRUSSELS SOUTH CHARLEROI AP, Belgium

WMO: 064490

Lat: 50.4542N

Lon: 4.4400E

Elev: 614

StdP: 14.37

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.0	23.6	14.2	11.6	21.5	18.5	14.2	25.4	29.1	48.0	26.6	47.5	6.9	60	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	85.4	68.9	81.4	67.2	77.7	65.3	70.4	82.0	68.4	78.8	66.6	75.6	8.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	98.9	76.7	64.5	93.0	73.2	63.0	88.0	71.1	34.7	82.5	32.9	79.1	31.5	75.9	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	20.6	18.3	DB	15.4	91.7	5.7	3.3	11.3	94.1	7.9	96.0	4.7	97.9	0.5	100.3	(4)
(5)			WB	14.6	73.9	5.7	2.5	10.5	75.7	7.1	77.2	3.9	78.6	-0.2	80.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	37.4	38.9	43.8	49.6	56.0	61.3	65.2	64.7	58.9	52.4	44.3	38.9	(6)
(7)		DBStd	11.80	7.44	7.24	6.54	6.87	6.53	6.04	6.10	5.57	5.33	6.35	6.43	7.26	(7)
(8)		HDD50	1605	392	312	209	89	17	0	0	0	1	48	188	349	(8)
(9)		HDD65	5307	855	730	658	463	288	145	74	73	198	393	620	810	(9)
(10)		CDD50	1979	1	2	17	77	204	339	473	456	267	122	18	3	(10)
(11)		CDD65	205	0	0	0	1	10	34	81	64	14	1	0	0	(11)
(12)		CDH74	1724	0	0	0	17	88	306	673	549	88	3	0	0	(12)
(13)		CDH80	480	0	0	0	0	14	76	203	174	13	0	0	0	(13)
(14)	Wind	WSAvg	8.7	10.3	10.0	9.6	8.3	8.0	7.6	7.5	7.0	7.6	8.7	9.1	10.2	(14)
(15)	Precipitation	PrecAvg	34.6	3.2	2.8	2.4	2.1	2.6	3.0	3.1	3.5	2.4	2.8	3.0	3.7	(15)
(16)		PrecMax	43.5	7.4	6.9	6.1	5.7	5.6	5.4	5.9	9.3	7.7	6.5	5.0	9.1	(16)
(17)		PrecMin	27.0	0.2	0.6	0.2	0.1	0.5	0.6	1.4	1.0	0.5	0.9	0.3	1.0	(17)
(18)		PrecStd	4.3	1.5	1.4	1.4	1.3	1.3	1.3	1.1	2.1	1.4	1.2	1.2	1.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.1	58.3	66.0	76.4	81.6	87.7	90.6	90.5	81.8	72.5	62.5	55.5	(19)
(20)			MCWB	51.2	48.2	53.0	60.5	66.5	69.8	69.2	70.5	66.8	62.1	55.9	52.3	(20)
(21)		2%	DB	51.8	53.5	61.0	70.5	75.9	81.5	85.4	84.8	76.3	67.7	58.0	53.1	(21)
(22)			MCWB	49.3	47.7	51.2	56.6	62.7	67.5	68.5	69.3	64.7	60.2	53.7	50.6	(22)
(23)		5%	DB	49.3	50.7	56.7	65.3	72.3	77.1	81.2	80.3	71.7	64.0	55.4	51.1	(23)
(24)			MCWB	46.9	47.1	49.6	54.3	60.5	65.1	66.9	67.0	62.8	58.5	52.4	49.0	(24)
(25)		10%	DB	47.2	48.5	53.4	60.9	68.0	73.1	77.2	75.6	67.9	61.1	53.1	48.8	(25)
(26)			MCWB	45.1	45.7	48.1	52.3	58.4	62.9	65.3	64.7	60.2	56.9	50.4	46.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	51.1	54.9	62.0	68.4	71.8	72.4	73.9	68.8	63.3	57.1	53.1	(27)
(28)			MCDWB	53.5	54.5	62.9	73.9	79.4	84.0	85.1	86.7	78.0	69.5	60.9	54.6	(28)
(29)		2%	WB	49.7	49.0	52.4	57.9	64.9	68.4	70.1	70.5	65.9	61.1	54.5	51.1	(29)
(30)			MCDWB	51.5	51.8	58.4	67.1	73.5	79.1	81.8	81.7	74.2	66.1	57.2	52.8	(30)
(31)		5%	WB	47.2	47.6	50.6	55.3	61.8	66.2	68.1	68.0	63.6	59.2	52.6	49.2	(31)
(32)			MCDWB	48.9	50.2	55.4	63.1	69.1	74.9	78.8	77.8	69.9	63.0	55.2	51.1	(32)
(33)		10%	WB	45.2	45.9	48.8	53.1	59.3	64.0	66.2	65.7	61.6	57.5	50.6	47.0	(33)
(34)			MCDWB	47.0	48.5	52.8	60.3	66.6	71.5	75.5	73.6	66.8	60.7	53.0	48.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.7	9.3	12.3	15.0	15.5	16.2	16.1	15.9	14.1	11.6	8.4	7.4	(35)
(36)			MCDBR	8.9	12.1	18.2	21.5	22.1	22.8	22.7	23.4	19.7	14.7	10.6	8.1	(36)
(37)			MCWBR	7.9	8.7	11.1	11.3	11.4	11.4	9.8	10.6	10.1	8.7	7.8	7.0	(37)
(38)		5% WB	MCDWB	8.6	10.0	15.1	19.0	19.4	20.4	20.3	21.2	17.2	13.4	9.4	7.8	(38)
(39)			MCWBR	8.1	8.1	10.0	10.8	11.2	11.2	9.9	10.9	9.4	8.6	7.6	7.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.318	0.340	0.372	0.408	0.418	0.416	0.417	0.409	0.380	0.356	0.331	0.315	(40)	
(41)		taud	2.415	2.379	2.288	2.203	2.215	2.243	2.240	2.286	2.353	2.420	2.416	2.385	(41)	
(42)		Ebn at Noon	227	251	261	264	266	267	265	261	256	241	218	207	(42)	
(43)		Edh at Noon	20	26	34	41	43	42	42	38	32	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	250	463	830	1314	1556	1700	1617	1376	1044	617	295	198	(44)	
(45)		RadStd	36	91	117	153	158	156	190	148	109	52	36	31	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-1.31	N/A	N/A	N/A	N/A
(47) Regional (78 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page